

AARHUS UNIVERSITY

DEPARTMENT OF MATHEMATICS

INTRODUCTION TO ALGEBRA 1

Handin 3

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September 15, 2026



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#MoreConsistency

Let a and b be natural numbers and $M \in \mathbb{N}$ be a common multiple of a and b .

Show that the product ab divides the product Ma and Mb .

From definition 3.8 [1] it holds that M is an integer that is divisible by both a and b , hence M can be defined as $a|M$ and $b|M$ and further be define per the start of Chapte 3 [1] as:

$$M = a \cdot q_1 \quad M = b \cdot q_2 \quad (1)$$

For $a|M$, multiply both sides by b and substitute M with (1)

$$a|M \quad (2)$$

$$= ab|Mb \quad (3)$$

$$= ab|(a \cdot q_1)b \quad (4)$$

$$= ab|(ab) \cdot q_1 \quad (5)$$

which should hold since q_1 is a valid integer that is not zero, hence it holds that $ab|Mb$, the same can be shown for $ab|Ma$ where it start woth mulplying by a and then substitute with $M = b \cdot q_2$ instead.

Show that the product ab divides the product $\gcd(a, b) \cdot M$.

Using bezouts identity theorem 3.3 [1] let

$$\gcd(a, b) = d = ax + by \quad (6)$$

take the product $\gcd(a, b) \cdot M$ and substitute using (6)

$$(ax + by) \cdot M = (Ma)x + (Mb)y$$

Then from earlier, substitute Ma and Mb with result from (2) to get

$$(Ma)x + (Mb)y = (ab \cdot q_1)x + (ab \cdot q_2)y = ab(q_1x + q_2y) \quad (7)$$

and since $(q_1x + q_2y) \in \mathbb{Z}$ per the definition of division (7) should atleast be divisible by ab hence $ab|\gcd(a, b) \cdot M$

Show that $\frac{ab}{\gcd(a, b)}$ divides M .

It was just shown that $ab|\gcd(a, b) \cdot M$ which implies $\gcd(a, b) \cdot M = (ab) \cdot q_3$ and since $\gcd(a, b) > 0$ and it's a whole number, it can be used to divide both sides to get:

$$M = \frac{ab}{\gcd(a, b)} \cdot q_3$$

which by the definition in the start of chapter 3 [1] shows that $\frac{ab}{\gcd(a, b)}|M$

Show that $\frac{ab}{\gcd(a,b)}$ is the least common multiple of a and b .

Let $M' = \frac{ab}{\gcd(a,b)}$ and reiterate that $M \in \mathbb{N}$ is a common multiple of a and b .

To show that $M' = \text{lcm}(a, b)$ from definition 3.8 [1] then it has to be shown that $a|M'$ and $b|M'$ which is to show $M' = a \cdot q_4$ and $M' = b \cdot q_5$ for some whole numbers $q_i \in \mathbb{W}$.

Take $M' = \frac{ab}{\gcd(a,b)}$ and separate a from the fraction and let $\frac{b}{\gcd(a,b)} = q_4$ this means we get the expression

$$M' = a \cdot q_4$$

which means $a|M'$, the same steps can be taken for b , so now it holds for both a and b that $a|M'$ and $b|M'$ so M' must be a common multiplier of a and b .

Now to show that M' is the least common multiple it should be enough to show that $M' \leq M$ since if $M' \leq M$ then M' is smaller than ANY other common multiple to a and b hence it must then be the least common multiple.

This should be trivial since it's already known that $\frac{ab}{\gcd(a,b)}|M = M'|M$ and since both $M', M > 0$ then it is implied by $M'|M$ that atleast $M' \leq M$.

In conclusion since M is any common multiple of a, b and $M' \leq M$ then

$$M' = \text{lcm}(a, b) = \frac{ab}{\gcd(a, b)}$$

References

- [1] Johan P. Hansen. *Tal og polynomier. Begreber, metoder, resultater, kodning og kryptografi*. Aarhus Universitetsforlag, 2018.